



Formulation of binary Darboux transformations for an integrable mKdV system with skew-symmetric matrix potentials

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ABSTRACT

The article is devoted to constructing a family of binary Darboux transformations for a skew-symmetric matrix integrable mKdV system. AKNS-type Lax pairs associated with the orthogonal Lie algebra $so(2m)$ are employed to systematically develop the proposed binary Darboux transformations. Starting from the zero background state, the application of these transformations generates explicit soliton solutions. A notable feature is that the M -matrix underlying the binary Darboux transformations is constructed within an extended framework in which eigenvalues coincide with adjoint eigenvalues, thereby yielding generalized Darboux transformations.

1. Introduction

Integrable models exhibit rich structures of coherent phenomena, such as solitons, breathers, lump waves, and rogue waves. Lax pairs serve as the starting point for the study of such systems and comprise two linear matrix spectral problems [1]. The inverse scattering transform [2] is developed within this framework to solve initial value problems, while the Riemann–Hilbert approach [3] provides an alternative method for analyzing their analytic properties based on complex analysis. In addition, Darboux transformations (DTs) [4,5] offer a direct technique for constructing explicit solutions from known ones.

Let $u = u(x, t)$ denote the potential in $(1+1)$ -dimensions and λ be the spectral parameter. An integrable model is associated with two linear spectral problems defining its Lax pair:

$$\phi_x = U\phi = U(u, \lambda)\phi, \quad \phi_t = V\phi = V(u, \lambda)\phi, \quad (1.1)$$

where U and V are square matrices. The solvability of these spectral problems is ensured by the compatibility condition $\phi_{xt} = \phi_{tx}$, which determines the integrable model. It is straightforward to verify that this compatibility condition is equivalent to the zero-curvature equation [6]:

$$U_t - V_x + [U, V] = 0. \quad (1.2)$$

The corresponding adjoint matrix spectral problems are given by

$$\psi_x = -\psi U, \quad \psi_t = -\psi V, \quad (1.3)$$

which also play a crucial role in constructing binary DTs. Their compatibility condition $\psi_{xt} = \psi_{tx}$ yields the same zero-curvature equation and therefore does not introduce any new constraint.

This paper considers an integrable modified Korteweg-de Vries (mKdV) system with two skew-symmetric matrix potentials p and q :

$$\begin{aligned} p_t &= \frac{1}{4}(p_{xxx} - 3pq p_x - 3p_x q p), \\ q_t &= \frac{1}{4}(q_{xxx} - 3q_x p q - 3q p q_x), \quad p^T = -p, \quad q^T = -q. \end{aligned} \quad (1.4)$$

This system is generated from the following Lax pair associated with the orthogonal Lie algebra $so(2m)$:

$$\begin{cases} U = \begin{bmatrix} \lambda I_m & p \\ q & -\lambda I_m \end{bmatrix}, \\ V = \lambda^3 \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix} + \lambda^2 \begin{bmatrix} 0 & p \\ q & 0 \end{bmatrix} - \frac{1}{2} \lambda \begin{bmatrix} pq & -p_x \\ q_x & -qp \end{bmatrix} \\ + \frac{1}{4} \begin{bmatrix} pq_x - p_x q & p_{xx} - 2pqp \\ q_{xx} - 2qpq & qp_x - q_x p \end{bmatrix}, \end{cases} \quad (1.5)$$

where p and q are skew-symmetric matrices of order m and I_m is the identity matrix of order m .

We focus on constructing binary DTs for this skew-symmetric matrix integrable mKdV system. A binary DT is implemented via a gauge

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transformation (see, e.g., [7]):

$$\phi' = S^+ \phi, \psi' = \psi S^-, S^+ = S^+(u, \lambda), S^- = S^-(u, \lambda), \quad (1.6)$$

where S^+ and S^- are mutual inverses and can be derived from a so-called M -matrix formulation (see, e.g., [7] for a general framework). The transformation is required to satisfy

$$\begin{cases} S_x^+(S^+)^{-1} + S^+ U(S^+)^{-1} = U(u', \lambda), \\ S_t^+(S^+)^{-1} + S^+ V(S^+)^{-1} = V(u', \lambda), \end{cases} \quad (1.7)$$

where the transformed potential u' is defined by

$$u' = f(u). \quad (1.8)$$

Accordingly, ϕ' and ψ' form a new eigenfunction and its adjoint satisfying

$$\phi'_x = U' \phi', \phi'_t = V' \phi', \quad (1.9)$$

and

$$\psi'_x = -\psi' U', \psi'_t = -\psi' V', \quad (1.10)$$

where

$$U' = U(u', \lambda), V' = V(u', \lambda). \quad (1.11)$$

Thus, the transformed eigenfunctions ϕ' and ψ' ensure that the resulting Lax pair (U', V') satisfies the same zero-curvature equation as the original, with u replaced by u' . Consequently, the transformed dependent variable u' yields a new solution to the integrable model determined by (U, V) .

In modern soliton theory, binary DTs have been widely applied to a variety of integrable models. Two representative examples are the nonlinear Schrödinger (NLS) equation and the mKdV equation (see, e.g., [8–10]). Determinant representations of n -fold binary DTs have been developed for the AKNS NLS equation and the Gerdjikov-Ivanov mKdV equation [11,12]. These binary DTs enable the generation of N -soliton solutions, including breathers, rogue waves, and lump waves as special cases. For Lax pairs with specific structures, it is also common to impose orthogonality conditions between eigenfunctions and adjoint eigenfunctions (see, e.g., [8]), particularly in situations where eigenvalues coincide with adjoint eigenvalues.

In this paper, we establish a class of binary DTs for the skew-symmetric matrix integrable mKdV system introduced above. The resulting transformations are constructed via an extended M -matrix generating procedure, which accommodates cases where eigenvalues coincide with adjoint eigenvalues. Applying the proposed transformations once yields explicit soliton solutions. The paper concludes by summarizing the main results and providing concluding remarks.

2. A skew-symmetric matrix integrable mKdV system

Let m be an arbitrary natural number. We consider the following coupled system of integrable mKdV equations:

$$\begin{cases} p_t = \frac{1}{4}(p_{xxx} - 3pq p_x - 3p_x q p), \\ q_t = \frac{1}{4}(q_{xxx} - 3q_x p q - 3q p q_x), \end{cases} \quad (2.1)$$

where p and q are two $m \times m$ skew-symmetric matrices. This system is generated from the Lax pair of matrix spectral problems:

$$\phi_x = U \phi, \phi_t = V \phi, \quad (2.2)$$

where U and V are associated with the orthogonal Lie algebra $so(2m)$ and are given by (1.5). The Lax pair can be written as

$$U = U(u, \lambda) = \lambda \Lambda + P, \quad (2.3)$$

and

$$V = V(u, \lambda) = \lambda^3 \Lambda + \lambda^2 P - \frac{1}{2} \lambda \Lambda (P^2 - P_x) + \frac{1}{4} ([P, P_x] + P_{xx} - 2P^3), \quad (2.4)$$

where $u = u(p, q)$ and

$$\Lambda = \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix}, P = \begin{bmatrix} 0 & p \\ q & 0 \end{bmatrix}. \quad (2.5)$$

The skew-symmetric conditions

$$p^T = -p, q^T = -q, \quad (2.6)$$

are equivalent to the potential reduction

$$\Sigma P \Sigma^{-1} = -P^T, \Sigma = \begin{bmatrix} 0 & I_m \\ I_m & 0 \end{bmatrix}, \quad (2.7)$$

which ensures that P , and hence U and V , belong to the orthogonal Lie algebra $so(2m)$.

Associated with the above spatial spectral matrix U , there exists an infinite number of commuting isospectral flows:

$$u_t = X^{[k]} = X^{[k]}(u) = \begin{bmatrix} 2b^{[k+1]} \\ -2c^{[k+1]} \end{bmatrix}, k \geq 0, \quad (2.8)$$

which are generated from the Lax operators $V^{[k]}$:

$$V^{[k]} = \sum_{l=0}^k \lambda^{k-l} W^{[l]}, W^{[l]} = \begin{bmatrix} a^{[l]} & b^{[l]} \\ c^{[l]} & d^{[l]} \end{bmatrix}. \quad (2.9)$$

Here, the Laurent series

$$W = \sum_{l \geq 0} W^{[l]} \lambda^{-l}, W^{[0]} = \Lambda, \quad (2.10)$$

is assumed to solve the stationary zero-curvature equation (see, e.g., [13,14]):

$$W_x = [U, W]. \quad (2.11)$$

The hierarchy is uniquely determined by imposing the normalization condition

$$W^{[l]}|_{u=0} = 0, l \geq 1. \quad (2.12)$$

In particular, the member corresponding to $k = 3$ coincides with the matrix integrable mKdV system (2.1), noting that

$$V = \lambda^3 W^{[0]} + \lambda^2 W^{[1]} + \lambda W^{[2]} + W^{[3]}. \quad (2.13)$$

The commutativity of these flows,

$$\llbracket X^{[k]}, X^{[l]} \rrbracket := X^{[k]'}(u)[X^{[l]}] - X^{[l]'}(u)[X^{[k]}] = 0, k, l \geq 0, \quad (2.14)$$

can be verified via the isospectral Lax operator algebra (see, e.g., [15]):

$$\llbracket V^{[k]}, V^{[l]} \rrbracket := V^{[k]'}(u)[X^{[l]}] - V^{[l]'}(u)[X^{[k]}] + [V^{[k]}, V^{[l]}] = 0, k, l \geq 0, \quad (2.15)$$

By imposing suitable reductions, the integrable mKdV system (2.1) reduces to several well-known coupled integrable models, thereby enlarging the category of multi-component mKdV-type systems studied previously (see, e.g., [16–20]).

In particular, under the reduction

$$p(t, x) = v(\mp \frac{t}{4}, x), q(t, x) = \mp v^\dagger(\mp \frac{t}{4}, x), \quad (2.16)$$

where $\dagger$ denotes the Hermitian conjugate, the system (2.1) reduces to the matrix focusing/defocusing mKdV equation:

$$v_t + 3vv^\dagger v_x + 3v_x v^\dagger v \pm v_{xxx} = 0, \quad (2.17)$$

where v is a square skew-symmetric matrix.

When

$$v = \begin{bmatrix} 0 & w \\ -w & 0 \end{bmatrix},$$

with w a scalar function, this matrix equation further reduces to the standard scalar focusing/defocusing mKdV equations:

$$w_t - 6|w|^2 w_x \pm w_{xxx} = 0, \quad (2.18)$$

which become

$$w_t - 6w^2 w_x \pm w_{xxx} = 0, \quad (2.19)$$

when w is real. These equations possess a variety of interesting exact solutions (see, e.g., [21–24]) and have important applications in nonlinear wave propagation (see, e.g., [25]).

3. Binary Darboux transformations

We consider the skew-symmetric matrix integrable mKdV system (2.1) together with its associated Lax pair defined in (2.2), where the matrices U and V are given by (2.3) and (2.4), respectively.

The objective of this section is to construct a class of binary DTs based on this Lax pair formulation. To incorporate both eigenfunctions and adjoint eigenfunctions, we develop an extended M -matrix framework for generating DTs for the matrix integrable mKdV system (2.1).

In the first subsection, we establish a binary DT formulation for the Lax pair associated with the general linear Lie algebra $gl(2m)$. In the second subsection, this construction is reduced to the case associated with the orthogonal Lie algebra $so(2m)$.

3.1. The general case associated with $gl(2m)$

Let u be a prescribed solution of the integrable mKdV system (2.1), with two unreduced matrix potentials p and q . Suppose that N is an arbitrary natural number, and consider a class of binary DTs constructed from N eigenfunction and adjoint eigenfunction pairs.

Assume that λ_i and $\hat{\lambda}_i$ ($1 \leq i \leq N$) are eigenvalues and adjoint eigenvalues, respectively, and that v_i and $\hat{v}_i$ are the corresponding eigenfunctions and the associated adjoint eigenfunctions. Then we have

$$v_{i,x} = U(u, \lambda_i)v_i, \quad v_{i,t} = V(u, \lambda_i)v_i, \quad 1 \leq i \leq N, \quad (3.1)$$

$$\hat{v}_{i,x} = -\hat{v}_i U(u, \hat{\lambda}_i), \quad \hat{v}_{i,t} = -\hat{v}_i V(u, \hat{\lambda}_i), \quad 1 \leq i \leq N. \quad (3.2)$$

We employ these eigenfunctions and adjoint eigenfunctions to establish a formulation for N -fold binary DTs. This framework enables the construction of new solutions of the matrix integrable mKdV system (2.1) from known ones.

To construct binary DTs, we introduce

$$\begin{cases} \Theta_{ij} := \frac{U(u, \lambda_j) - U(u, \hat{\lambda}_i)}{\lambda_j - \hat{\lambda}_i} = \Lambda, \\ \Xi_{ij} := \frac{V(u, \lambda_j) - V(u, \hat{\lambda}_i)}{\lambda_j - \hat{\lambda}_i} = (\lambda_j^2 + \lambda_j \hat{\lambda}_i + \hat{\lambda}_i^2)\Lambda + (\lambda_j + \hat{\lambda}_i)P - \frac{1}{2}\Lambda(P^2 - P_x). \end{cases} \quad (3.3)$$

A crucial step in constructing the N -fold binary DTs is the introduction of an $N \times N$ matrix $M = (m_{ij})$, based on these quantities. To encode the interactions between the eigenfunctions and adjoint eigenfunctions (see, e.g., [10,26]), we define

$$M = (m_{ij})_{N \times N}, \quad m_{ij} = \begin{cases} \frac{\hat{v}_i v_j}{\lambda_j - \hat{\lambda}_i}, & \text{if } \lambda_j \neq \hat{\lambda}_i, \\ m_{ij}^c(x, t), & \text{if } \lambda_j = \hat{\lambda}_i, \end{cases} \quad \text{where } 1 \leq i, j \leq N, \quad (3.4)$$

where v_i and $\hat{v}_i$ are the eigenfunctions and adjoint eigenfunctions introduced in (3.1) and (3.2), respectively. The entries $m_{ij}^c(x, t)$ correspond to the coinciding-eigenvalue case $\lambda_j = \hat{\lambda}_i$, and are defined by

$$m_{ij,x}^c = \hat{v}_i \Theta_{ij} v_j, \quad \text{if } \lambda_j = \hat{\lambda}_i, \quad (3.5)$$

and

$$m_{ij,t}^c = \hat{v}_i \Xi_{ij} v_j, \quad \text{if } \lambda_j = \hat{\lambda}_i, \quad (3.6)$$

for $1 \leq i, j \leq N$.

The M -matrix plays a central role in constructing the higher-rank projector that defines the binary Darboux matrices. In particular, it encompasses generalized DTs in which eigenvalues may coincide with adjoint eigenvalues. One particular case is obtained by setting

$$m_{ij}^c = 0, \quad \text{if } \lambda_j = \hat{\lambda}_i. \quad (3.7)$$

This requires the orthogonality conditions (see, e.g., [7,8]):

$$\hat{v}_i \Theta_{ij} v_j = 0, \quad \hat{v}_i \Xi_{ij} v_j = 0, \quad \text{if } \lambda_j = \hat{\lambda}_i. \quad (3.8)$$

As soon as the $N \times N$ matrix M defined in (3.4) is invertible, we can formulate the N -fold binary DTs as follows. More precisely, the Darboux matrices are defined by

$$S^+ = S^+(\lambda) = I_{2m} - \sum_{i,j=1}^N \frac{v_i (M^{-1})_{ij} \hat{v}_j}{\lambda - \hat{\lambda}_i}, \quad S^- = S^-(\lambda) = I_{2m} + \sum_{i,j=1}^N \frac{v_i (M^{-1})_{ij} \hat{v}_j}{\lambda - \lambda_i}, \quad (3.9)$$

together with their leading-order terms at infinity:

$$S_1^\pm = \lim_{\lambda \rightarrow \infty} [\lambda(S^\pm(\lambda) - I_{2m})]. \quad (3.10)$$

In the above formulation, v_i and $\hat{v}_i$ are the eigenfunctions and adjoint eigenfunctions defined in (3.1) and (3.2), respectively.

Based on the general theory of binary DTs, we need to verify the following two gauge conditions:

$$S_x^+ = U' S^+ - S^+ U, \quad (3.11)$$

and

$$S_t^+ = V' S^+ - S^+ V. \quad (3.12)$$

Here, the resulting Lax pair (U', V') is defined by

$$U' = \lambda \Lambda + P', \quad V' = \lambda^3 \Lambda + \lambda^2 P' - \frac{1}{2} \lambda \Lambda (P'^2 - P_x') + \frac{1}{4} ([P', P_x'] + P_{xx}' - 2P'^3), \quad (3.13)$$

where the transformed potential matrix P' is given by

$$P' = \begin{bmatrix} 0 & p' \\ q' & 0 \end{bmatrix} = P + S_1^+ \Lambda + \Lambda S_1^-. \quad (3.14)$$

Thus, constructing a family of binary DTs for the matrix integrable mKdV system (2.1) reduces to verifying the two conditions (3.11) and (3.12).

We observe that, under the non-degeneracy conditions

$$\begin{cases} \{\lambda_i \mid 1 \leq i \leq N\} \cap \{\hat{\lambda}_i \mid 1 \leq i \leq N\} = \emptyset; \\ \lambda_i \neq \lambda_j, \quad 1 \leq i, j \leq N; \\ \hat{\lambda}_i \neq \hat{\lambda}_j, \quad 1 \leq i, j \leq N; \end{cases} \quad (3.15)$$

the expression (3.9) reduces to the standard N -fold Darboux transformation (DT) formulation, corresponding to Darboux matrices of the classical type, which has been extensively studied in the literature (see, e.g., [4,5,26–30]). If some eigenvalues coincide with adjoint eigenvalues, the construction of the matrix M requires additional care, leading to more intricate solution structures [10]. Such cases arise naturally, for instance, in spectral problems associated with reductions, including those related to orthogonal Lie algebras.

We are now in a position to state the following two propositions, in a manner analogous to many related settings (see, e.g., [8,10]).

Using the expressions for the Darboux matrices S^+ and S^- in (3.9), one can obtain their partial fraction decompositions, as summarized below.

Proposition 3.1. Let S^+ and S^- be the Darboux matrices defined by (3.9). Then their partial fraction decompositions are given by

$$S^+ = I_{2m} - \sum_{i=1}^N \frac{v_i^M \hat{v}_i}{\lambda - \hat{\lambda}_i}, \quad S^- = I_{2m} + \sum_{i=1}^N \frac{v_i \hat{v}_i^M}{\lambda - \hat{\lambda}_i} \quad (3.16)$$

where the vectors v_i^M and $\hat{v}_i^M$ are defined via the $N \times N$ matrix M by

$$\begin{cases} (v_1^M, v_2^M, \dots, v_N^M)M = (v_1, v_2, \dots, v_N), \\ M((\hat{v}_1^M)^T, (\hat{v}_2^M)^T, \dots, (\hat{v}_N^M)^T)^T = (\hat{v}_1^T, \hat{v}_2^T, \dots, \hat{v}_N^T)^T. \end{cases} \quad (3.17)$$

From the above proposition, we immediately obtain the following result.

Proposition 3.2. For the Darboux matrices S^+ and S^- defined in (3.9), the following properties hold:

$$S^+(\lambda_i)v_i = 0, \quad \hat{v}_i S^-(\hat{\lambda}_i) = 0, \quad 1 \leq i \leq N, \quad (3.18)$$

provided that $S^+(\lambda_i)$ and $S^-(\hat{\lambda}_i)$ are well-defined. Moreover,

$$S^+(\lambda)S^-(\lambda) = I_{2m}, \quad S_1^+ = -S_1^-, \quad (3.19)$$

holds under the orthogonality condition

$$\hat{v}_i v_j = 0, \quad \text{if } \lambda_j = \hat{\lambda}_i, \quad \text{where } 1 \leq i, j \leq N, \quad (3.20)$$

is satisfied.

The condition (3.20) is required in the case of coincidence between eigenvalues and adjoint eigenvalues. In particular, the first identity in (3.19) implies that $S^+(\lambda)$ and $S^-(\lambda)$ are mutual inverses.

For brevity, we introduce the block vectors:

$$v = (v_1, v_2, \dots, v_N), \quad \hat{v} = (\hat{v}_1^T, \hat{v}_2^T, \dots, \hat{v}_N^T)^T, \quad (3.21)$$

together with the diagonal matrices

$$\begin{aligned} F &= \text{diag}\left(\frac{1}{\lambda - \lambda_1}, \frac{1}{\lambda - \lambda_2}, \dots, \frac{1}{\lambda - \lambda_N}\right), \\ \hat{F} &= \text{diag}\left(\frac{1}{\lambda - \hat{\lambda}_1}, \frac{1}{\lambda - \hat{\lambda}_2}, \dots, \frac{1}{\lambda - \hat{\lambda}_N}\right). \end{aligned} \quad (3.22)$$

Then the Darboux matrices admit the compact representations

$$\begin{cases} S^+ = I_{2m} - v M^{-1} \hat{F} \hat{v}, \\ S^- = I_{2m} + v F M^{-1} \hat{v}. \end{cases} \quad (3.23)$$

One can directly verify the identity

$$\hat{v} v = \hat{F}^{-1} M - M F^{-1}, \quad (3.24)$$

which holds entrywise. Using this relation, a straightforward computation shows that

$$S^+ S^- = I_{2m} - v M^{-1} \hat{F} \hat{v} + v F M^{-1} \hat{v} - v M^{-1} \hat{F} \hat{v} v F M^{-1} \hat{v} = I_{2m}.$$

Taking the expansion at infinity, we obtain

$$\begin{cases} S_1^+ = - \sum_{i,j=1}^N (M^{-1})_{ij} v_i \hat{v}_j = -v M^{-1} \hat{v}, \\ S_1^- = \sum_{i,j=1}^N (M^{-1})_{ij} v_i \hat{v}_j = v M^{-1} \hat{v}, \end{cases} \quad (3.25)$$

which immediately implies $S_1^+ = -S_1^-$.

We are now in a position to present a class of general binary DTs for the matrix integrable mKdV system (2.1), without imposing skew-symmetric reductions on p and q .

Theorem 3.1. Let v_i and $\hat{v}_i$ ($1 \leq i \leq N$) be defined by (3.1) and (3.2), respectively, and let the M -matrix be given by (3.4), with the entries m_{ij}^c defined via (3.5), (3.6), and (3.3). Assume that the orthogonality condition (3.20) holds. Then the matrix integrable mKdV system (2.1) admits the binary DT:

$$\phi' = S^+ \phi, \quad \psi' = \psi S^-, \quad P' = P + S_1^+ \Lambda + \Lambda S_1^-, \quad (3.26)$$

where $S^\pm$ and $S_1^\pm$ are given by (3.9) and (3.10), or equivalently by (3.23) and (3.25), respectively.

Proof. Let U' and V' be defined as in (3.13), with P' given by (3.14). The proof reduces to verifying the spatial and temporal compatibility conditions (3.11) and (3.12).

We first verify the spatial condition (3.11). From the spectral problems, we have

$$v_x = \Lambda v D + P v, \quad \hat{v}_x = -(\hat{D} \hat{v} \Lambda + \hat{v} P), \quad (3.27)$$

where

$$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N), \quad \hat{D} = \text{diag}(\hat{\lambda}_1, \hat{\lambda}_2, \dots, \hat{\lambda}_N). \quad (3.28)$$

Using the definition of M in (3.4), together with (3.5) and (3.6), we obtain

$$M_x = \hat{v} \Theta v, \quad \Theta = (\Theta_{ij})_{N \times N} = (\Lambda)_{N \times N}. \quad (3.29)$$

With the compact representations (3.23) and (3.25) for S^+ and $S_1^\pm$, a direct computation verifies the spatial condition (3.11). In this calculation, we use the identities

$$\hat{D} = -\hat{F}^{-1} + \lambda I_N, \quad D = -F^{-1} + \lambda I_N, \quad (3.30)$$

together with (3.24). Hence, the spatial condition is satisfied.

Next, we verify the temporal condition (3.12). We have

$$\begin{cases} v_t = \Lambda v D^3 + P v D^2 + W^{[2]} v D + W^{[3]} v, \\ \hat{v}_t = -(\hat{D}^3 \hat{v} \Lambda + \hat{D}^2 \hat{v} P + \hat{D} \hat{v} W^{[2]} + \hat{v} W^{[3]}), \end{cases} \quad (3.31)$$

where

$$W^{[2]} = -\frac{1}{2} \Lambda (P^2 - P_x), \quad W^{[3]} = \frac{1}{4} ([P, P_x] + P_{xx} - 2P^3), \quad (3.32)$$

and

$$M_t = \hat{v} \Xi v, \quad \Xi = (\Xi_{ij})_{N \times N} = ((\lambda_j^2 + \lambda_j \hat{\lambda}_i + \hat{\lambda}_i^2) \Lambda + (\lambda_j + \hat{\lambda}_i) P - \frac{1}{2} \Lambda (P^2 - P_x))_{N \times N}. \quad (3.33)$$

Equivalently, Ξ can be written in the compact form

$$\Xi = \hat{D}^2 \Lambda + \hat{D} \Lambda D + \Lambda D^2 + \hat{D} P + P D - \frac{1}{2} (\Lambda (P^2 - P_x))_{N \times N}. \quad (3.34)$$

A direct (though lengthy) computation using the compact forms of S^+ and S_1^+ , together with

$$P' = P + S_1^+ \Lambda - \Lambda S_1^+ = P - v M^{-1} \hat{v} \Lambda + \Lambda v M^{-1} \hat{v},$$

and the identities (3.30) and (3.24), shows that the temporal condition (3.12) is also satisfied.

Therefore, both compatibility conditions hold, and the binary DT is established. $\square$

3.2. The skew-symmetric case associated with $so(2m)$

The transformed potential matrix P' defined by (3.14) must satisfy the orthogonal symmetry condition associated with $so(2m)$:

$$\Sigma P' \Sigma^{-1} = -(P')^T, \quad (3.35)$$

where Σ is defined in (2.7). Equivalently, using the binary DT in (3.26) together with (3.19), this reduction condition can be reformulated as

$$\Sigma [S_1^+, \Lambda] \Sigma^{-1} = [(S_1^+)^T, \Lambda], \quad (3.36)$$

where S_1^+ is given explicitly by (3.25).

To this end, we choose the adjoint eigenvalues as

$$\hat{\lambda}_i = \lambda_i, \quad 1 \leq i \leq N, \quad (3.37)$$

and define the adjoint eigenfunctions according to the symmetry of U and V :

$$\hat{v}_i = v_i^T \Sigma^T = v_i^T \Sigma, \quad 1 \leq i \leq N. \quad (3.38)$$

We impose the orthogonality condition

$$\hat{v}_j v_i = 0, \quad \text{when } \lambda_j = \lambda_i, \quad 1 \leq i, j \leq N, \quad (3.39)$$

to ensure that the two Darboux matrices S^+ and S^- are mutual inverses.

For $\hat{\lambda}_j \neq \lambda_i$, we compute

$$m_{ij} = \frac{\hat{v}_i v_j}{\lambda_j - \lambda_i} = \frac{v_i^T \Sigma v_j}{\lambda_j - \lambda_i} = -\frac{v_i^T \Sigma v_j}{\lambda_i - \lambda_j} = -\frac{\hat{v}_j v_i}{\lambda_i - \lambda_j} = -m_{ji},$$

showing that the off-diagonal entries are skew-symmetric.

In the resonant case $\hat{\lambda}_j = \lambda_i$, $1 \leq i, j \leq N$, we observe

$$(m_{ij}^c + m_{ji}^c)_x = (m_{ij}^c + m_{ji}^c)_t = 0, \quad (3.40)$$

so that we can choose $M^T = -M$.

Then, from (3.25), it follows that

$$\Sigma S_1^+ \Sigma^{-1} = -(S_1^+)^T, \quad (3.41)$$

and together with

$$\Sigma \Lambda \Sigma^{-1} = -\Lambda, \quad (3.42)$$

this implies (3.36), and consequently the reduction property (3.35) for the transformed potential P' .

The following theorem summarizes the main result.

Theorem 3.2. Let the adjoint eigenvalues $\{\hat{\lambda}_i \mid 1 \leq i \leq N\}$ and the adjoint eigenfunctions $\{\hat{v}_i \mid 1 \leq i \leq N\}$ be defined by (3.37) and (3.38), respectively. Let M be given by (3.4), and assume that whenever $\hat{\lambda}_j = \lambda_i$ for $1 \leq i, j \leq N$, the corresponding entries m_{ij}^c of M are determined by (3.5) and (3.6), satisfying

$$m_{ij}^c = -m_{ji}^c, \text{ when } \hat{\lambda}_j = \lambda_i, \quad 1 \leq i, j \leq N, \quad (3.43)$$

so that $M^T = -M$. Moreover, if the orthogonality conditions (3.39) are satisfied, then the transformation (3.26) yields a binary DT for the skew-symmetric matrix integrable mKdV system (2.1).

3.3. An illustrative example

We note that N must be even to ensure that M is skew-symmetric and invertible. Let us now present an example with $m = 4$ and $N = 2$, starting from the zero seed potential. The resulting mKdV system is given by

$$\begin{cases} p_{12,t} = \frac{1}{4}p_{12,xxx} + \frac{3}{4}(2p_{12}q_{12} + p_{13}q_{13} + p_{14}q_{14} + p_{23}q_{23} + p_{24}q_{24})p_{12,x} \\ \quad + \frac{3}{4}(p_{12}q_{13} + p_{24}q_{34})p_{13,x} + \frac{3}{4}(p_{12}q_{14} - p_{23}q_{34})p_{14,x} \\ \quad + \frac{3}{4}(p_{12}q_{23} - p_{14}q_{34})p_{23,x} + \frac{3}{4}(p_{12}q_{24} + p_{13}q_{34})p_{24,x}, \\ p_{13,t} = \frac{1}{4}p_{13,xxx} + \frac{3}{4}(p_{12}q_{12} + 2p_{13}q_{13} + p_{14}q_{14} + p_{23}q_{23} + p_{34}q_{34})p_{13,x} \\ \quad + \frac{3}{4}(p_{13}q_{12} + p_{34}q_{24})p_{12,x} + \frac{3}{4}(p_{13}q_{14} + p_{23}q_{24})p_{14,x} \\ \quad + \frac{3}{4}(p_{13}q_{23} + p_{14}q_{24})p_{23,x} + \frac{3}{4}(p_{12}q_{24} + p_{13}q_{34})p_{34,x}, \\ p_{14,t} = \frac{1}{4}p_{14,xxx} + \frac{3}{4}(p_{12}q_{12} + p_{13}q_{13} + 2p_{14}q_{14} + p_{24}q_{24} + p_{34}q_{34})p_{14,x} \\ \quad + \frac{3}{4}(p_{14}q_{12} - p_{34}q_{23})p_{12,x} + \frac{3}{4}(p_{14}q_{13} + p_{24}q_{23})p_{13,x} \\ \quad + \frac{3}{4}(p_{13}q_{23} + p_{14}q_{24})p_{24,x} - \frac{3}{4}(p_{12}q_{23} - p_{14}q_{34})p_{34,x}, \\ p_{23,t} = \frac{1}{4}p_{23,xxx} + \frac{3}{4}(p_{12}q_{12} + p_{13}q_{13} + 2p_{23}q_{23} + p_{24}q_{24} + p_{34}q_{34})p_{23,x} \\ \quad + \frac{3}{4}(p_{23}q_{12} - p_{34}q_{14})p_{12,x} + \frac{3}{4}(p_{23}q_{13} + p_{24}q_{14})p_{13,x} \\ \quad + \frac{3}{4}(p_{13}q_{14} + p_{23}q_{24})p_{24,x} - \frac{3}{4}(p_{12}q_{14} - p_{23}q_{34})p_{34,x}, \\ p_{24,t} = \frac{1}{4}p_{24,xxx} + \frac{3}{4}(p_{12}q_{12} + p_{14}q_{14} + p_{23}q_{23} + 2p_{24}q_{24} + p_{34}q_{34})p_{24,x} \\ \quad + \frac{3}{4}(p_{24}q_{12} + p_{34}q_{13})p_{12,x} + \frac{3}{4}(p_{23}q_{13} + p_{24}q_{14})p_{13,x} \\ \quad + \frac{3}{4}(p_{14}q_{13} + p_{24}q_{23})p_{23,x} + \frac{3}{4}(p_{12}q_{13} + p_{24}q_{34})p_{34,x}, \\ p_{34,t} = \frac{1}{4}p_{34,xxx} + \frac{3}{4}(p_{13}q_{13} + p_{14}q_{14} + p_{23}q_{23} + p_{24}q_{24} + 2p_{34}q_{34})p_{34,x} \\ \quad + \frac{3}{4}(p_{24}q_{12} + p_{34}q_{13})p_{13,x} - \frac{3}{4}(p_{23}q_{12} - p_{34}q_{14})p_{14,x} \\ \quad - \frac{3}{4}(p_{14}q_{12} - p_{34}q_{23})p_{23,x} + \frac{3}{4}(p_{13}q_{12} + p_{34}q_{24})p_{24,x}; \end{cases}$$

and

$$\begin{cases} q_{12,t} = \frac{1}{4}q_{12,xxx} + \frac{3}{4}(2p_{12}q_{12} + p_{13}q_{13} + p_{14}q_{14} + p_{23}q_{23} + p_{24}q_{24})q_{12,x} \\ \quad + \frac{3}{4}(p_{13}q_{12} + p_{34}q_{24})q_{13,x} + \frac{3}{4}(p_{14}q_{12} - p_{34}q_{23})q_{14,x} \\ \quad + \frac{3}{4}(p_{23}q_{12} - p_{34}q_{14})q_{23,x} + \frac{3}{4}(p_{24}q_{12} + p_{34}q_{13})q_{24,x}, \\ q_{13,t} = \frac{1}{4}q_{13,xxx} + \frac{3}{4}(p_{12}q_{12} + 2p_{13}q_{13} + p_{14}q_{14} + p_{23}q_{23} + p_{34}q_{34})q_{13,x} \\ \quad + \frac{3}{4}(p_{12}q_{13} + p_{24}q_{34})q_{12,x} + \frac{3}{4}(p_{14}q_{13} + p_{24}q_{23})q_{14,x} \\ \quad + \frac{3}{4}(p_{23}q_{13} + p_{24}q_{14})q_{23,x} + \frac{3}{4}(p_{24}q_{12} + p_{34}q_{13})q_{34,x}, \\ q_{14,t} = \frac{1}{4}q_{14,xxx} + \frac{3}{4}(p_{12}q_{12} + p_{13}q_{13} + 2p_{14}q_{14} + p_{24}q_{24} + p_{34}q_{34})q_{14,x} \\ \quad + \frac{3}{4}(p_{12}q_{14} - p_{23}q_{34})q_{12,x} + \frac{3}{4}(p_{13}q_{14} + p_{23}q_{24})q_{13,x} \\ \quad + \frac{3}{4}(p_{23}q_{13} + p_{24}q_{14})q_{24,x} - \frac{3}{4}(p_{23}q_{12} - p_{34}q_{14})q_{34,x}, \\ q_{23,t} = \frac{1}{4}q_{23,xxx} + \frac{3}{4}(p_{12}q_{12} + p_{13}q_{13} + 2p_{23}q_{23} + p_{24}q_{24} + p_{34}q_{34})q_{23,x} \\ \quad + \frac{3}{4}(p_{12}q_{23} - p_{14}q_{34})q_{12,x} + \frac{3}{4}(p_{13}q_{23} + p_{14}q_{24})q_{13,x} \\ \quad + \frac{3}{4}(p_{14}q_{13} + p_{24}q_{23})q_{24,x} - \frac{3}{4}(p_{14}q_{12} - p_{34}q_{23})q_{34,x}, \\ q_{24,t} = \frac{1}{4}q_{24,xxx} + \frac{3}{4}(p_{12}q_{12} + p_{14}q_{14} + p_{23}q_{23} + 2p_{24}q_{24} + p_{34}q_{34})q_{24,x} \\ \quad + \frac{3}{4}(p_{12}q_{24} + p_{13}q_{34})q_{12,x} + \frac{3}{4}(p_{13}q_{23} + p_{14}q_{24})q_{14,x} \\ \quad + \frac{3}{4}(p_{13}q_{14} + p_{23}q_{24})q_{23,x} + \frac{3}{4}(p_{13}q_{12} + p_{34}q_{24})q_{34,x}, \\ q_{34,t} = \frac{1}{4}q_{34,xxx} + \frac{3}{4}(p_{13}q_{13} + p_{14}q_{14} + p_{23}q_{23} + p_{24}q_{24} + 2p_{34}q_{34})q_{34,x} \\ \quad + \frac{3}{4}(p_{12}q_{24} + p_{13}q_{34})q_{13,x} - \frac{3}{4}(p_{12}q_{23} - p_{14}q_{34})q_{14,x} \\ \quad - \frac{3}{4}(p_{12}q_{14} - p_{23}q_{34})q_{23,x} + \frac{3}{4}(p_{12}q_{13} + p_{24}q_{34})q_{24,x}; \end{cases}$$

where the two potentials are assumed to be $p = (p_{ij})_{4 \times 4}$ and $q = (q_{ij})_{4 \times 4}$. The eigenfunctions take the form

$$v_i = (e^{\lambda_i x + \hat{\lambda}_i^3 t} \mu_1^{[i]T}, e^{-\lambda_i x - \hat{\lambda}_i^3 t} \mu_2^{[i]T})^T, \quad 1 \leq i \leq 2, \quad (3.44)$$

and the corresponding adjoint eigenfunctions are given by

$$\hat{v}_i = v_i^T \Sigma, \quad 1 \leq i \leq 2. \quad (3.45)$$

Here $\mu_j^{[i]} \in \mathbb{C}^4$, $1 \leq i, j \leq 2$, are constant column vectors.

Since $\hat{\lambda}_i = \lambda_i$, $1 \leq i \leq 2$, the orthogonality conditions

$$\hat{v}_i v_i = v_i^T \Sigma v_i = 0, \quad 1 \leq i \leq 2, \quad (3.46)$$

must be satisfied, which is equivalent to

$$\mu_1^{[i]T} \mu_2^{[i]} = 0, \quad 1 \leq i \leq 2. \quad (3.47)$$

If we choose $m_{11}^c = m_{22}^c = 0$, two additional sets of orthogonality conditions arise. The first,

$$\hat{v}_i \Lambda v_i = 0, \quad 1 \leq i \leq 2, \quad (3.48)$$

is automatically satisfied. The second,

$$3\lambda_i^2 \hat{v}_i \Lambda v_i = 0, \quad 1 \leq i \leq 2, \quad (3.49)$$

follows directly from (3.48). Hence, only the conditions in (3.47) need to be imposed.

A direct computation using

$$p' = [S_1^+, \Lambda]_{12} = -2(S_1^+)_{12}, \quad q' = [S_1^+, \Lambda]_{21} = 2(S_1^+)_{21},$$

yields

$$p'_{ij} = -p'_{ji} = \frac{2(\lambda_1 - \lambda_2)(\mu_{1,1,i} \mu_{2,1,j} - \mu_{1,1,j} \mu_{2,1,i})}{\mu_2^{[1]T} \mu_1^{[2]} e^{-2(\lambda_1 x + \lambda_1^3 t)} + \mu_1^{[1]T} \mu_2^{[2]} e^{-2(\lambda_2 x + \lambda_2^3 t)}}, \quad 1 \leq i, j \leq 4, \quad (3.50)$$

and

$$q'_{ij} = -q'_{ji} = -\frac{2(\lambda_1 - \lambda_2)(\mu_{1,2,i}\mu_{2,2,j} - \mu_{1,2,j}\mu_{2,2,i})}{\mu_2^{[1]T}\mu_1^{[2]}e^{2(\lambda_2 x + \lambda_1^2 t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{2(\lambda_1 x + \lambda_2^2 t)}}, \quad 1 \leq i, j \leq 4. \quad (3.51)$$

Here we assume

$$p' = (p'_{ij})_{4 \times 4}, \quad q' = (q'_{ij})_{4 \times 4}, \quad \mu_s^{[r]} = (\mu_{r,s,1}, \mu_{r,s,2}, \mu_{r,s,3}, \mu_{r,s,4})^T, \quad 1 \leq r, s \leq 2. \quad (3.52)$$

Under the conditions in (3.47), these expressions yield explicit solutions of the matrix integrable mKdV system with skew-symmetric matrix potentials in (2.1).

When taking the eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = 2,$$

and the parameters

$$\begin{cases} \mu_{1,1,1} = 1, \mu_{1,1,2} = 2, \mu_{1,1,3} = 3, \mu_{1,1,4} = \frac{1}{4}, \\ \mu_{1,2,1} = -23, \mu_{1,2,2} = 3, \mu_{1,2,3} = 5, \mu_{1,2,4} = 8, \\ \mu_{2,1,1} = 3, \mu_{2,1,2} = 4, \mu_{2,1,3} = 5, \mu_{2,1,4} = 6, \\ \mu_{2,2,1} = -\frac{74}{3}, \mu_{2,2,2} = 7, \mu_{2,2,3} = 8, \mu_{2,2,4} = 1, \end{cases}$$

we obtain the following particular solutions:

$$p'_{ij} = -p'_{ji} = \frac{c_{ij}^{(p)} e^{3x+9t}}{192 e^{x+7t} + 163 e^{-x-7t}}, \quad 1 \leq i < j \leq 4,$$

where the coefficients are

$$c_{12}^{(p)} = 48, \quad c_{13}^{(p)} = 96, \quad c_{14}^{(p)} = -126, \quad c_{23}^{(p)} = 48, \quad c_{24}^{(p)} = -264, \quad c_{34}^{(p)} = -402.$$

Similarly,

$$q'_{ij} = -q'_{ji} = \frac{c_{ij}^{(q)} e^{-3x-9t}}{192 e^{x+7t} + 163 e^{-x-7t}}, \quad 1 \leq i < j \leq 4,$$

with coefficients

$$c_{12}^{(q)} = -2088, \quad c_{13}^{(q)} = -1456, \quad c_{14}^{(q)} = 4184, \quad c_{23}^{(q)} = -264, \quad c_{24}^{(q)} = -1272, \quad c_{34}^{(q)} = -1416.$$

4. Concluding remarks

This paper studies binary DTs for the integrable mKdV system with two skew-symmetric matrix potentials, based on an extended M -matrix generating scheme. The resulting DT framework plays a central role in analyzing the integrable dynamics of the matrix mKdV model. Starting from the vacuum state, iterative applications of these binary DTs produce a broad class of explicit exact soliton solutions.

The example presented here is associated with the orthogonal Lie algebra $so(2m)$ and highlights the diversity of coupled nonlinear integrable systems, demonstrating both the practical effectiveness of the approach and the richness of the underlying integrability structures. In particular, the M -matrix must be extended to accommodate cases in which eigenvalues coincide with adjoint eigenvalues, for which the classical M -matrix formulation is no longer applicable.

By examining the eigenvalues of the corresponding spectral problems, soliton-type solutions can be classified into categories such as negatons, positons, and complexitons, and further divided into breathers, solitons, rogue waves, kinks, anti-kinks, lump waves, and diverse mixed interaction solutions. Some interaction solutions may even involve arbitrary functions. Specific solutions can also be constructed via wave-number reductions of general soliton families (see, e.g., [31,32]). In particular, lump wave solutions have been widely investigated in both linear and nonlinear settings (see, e.g., [33–37]).

Furthermore, increasing the number of dependent variables or including higher powers of the auxiliary spectral parameter in the spatial spectral problem leads to larger and more intricate integrable systems

(see, e.g., [38,39]), among which integrable couplings provide representative examples (see, e.g., [40]). Such extensions—especially those involving nonlocal structures—introduce additional analytical challenges. Nevertheless, the classification of these generalized systems and the study of their dynamical behavior are expected to deepen our understanding of the underlying mechanisms governing nonlinear dynamics and integrable structures in mathematical physics [41,42].

Declaration of competing interest

The author declares that there is no known competing interest that could have appeared to influence this work.

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Data availability

No data was used for the research described in the article.

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